How to Build a Time Series Agent

Reuse what is general · adapt what is temporal · rebuild what is task-bound

Key claim: Align model capability, harness responsibility and evaluation

Scenario & data

series · context · task request

LLM

ALIGN

interprets temporal evidence and selects actions

- Represent numbers for the language model
- Reason over temporal evidence
- Learn which workflow action to take

→ Not enough alone: it acts through the harness

action

General Harness

REUSE

domain-independent runtime

- Interaction & memory
- Tools & execution
- Planning & coordination
- Feedback & self-improvement

→ Carry it over from general Agents

observation

feedback: revise

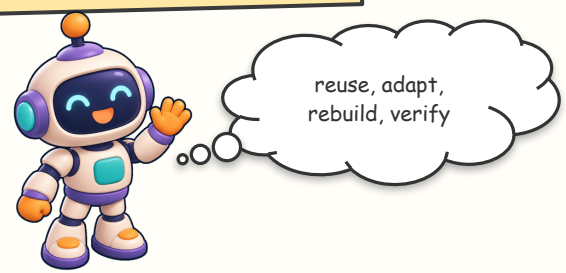
Evaluation

VERIFY

tests the complete workflow

- Process validity
- Temporal validity
- Task performance
- Decision utility

→ Valid run ≠ accurate result ≠ useful decision



temporal operations

General Time Series Harness

ADAPT

temporal semantics shared across tasks

- Timestamps, units, missingness
- Windows, resampling, decomposition
- Only information available at t_0
- Chronology guards & validators

→ Add time where a generic runtime is ambiguous

Task-Specific Harness

REBUILD

procedures bound to one task contract

- Forecasting
- Synthesis & augmentation
- Anomaly diagnosis
- Decision support

→ Rebuild around output, objective and actions

shared temporal capabilities

candidate output

Accepted output

accept / terminate

history: observed and available to the Agent

anomaly

future: forecast horizon, not yet observed

decision

t_0 information cutoff

time